TERAS ACADEMY

MASTER ARCHITECTURE

v1.0

Master Reference & Continuity Document

Menerjemahkan Kompleksitas Pembelajaran Menjadi Sistem Kapabilitas yang Dapat Dibuktikan

Institution	TERAS Institute
Document	TERAS Academy Master Architecture
Version	v1.0
Status	MASTER / CONTROLLED BASELINE

1. Document Control

Dokumen ini menjadi sumber acuan utama untuk arsitektur TERAS Academy. Dokumen turunannya wajib konsisten dengan struktur, prinsip, baseline, dan batasan yang ditetapkan di sini.

Elemen	Isi	Status	Catatan
Document ID	TA-MA-001	Locked	Master architecture
Version	1.0	Locked	Baseline awal
Owner	TERAS Academy / TERAS Institute	Locked	Institutional owner
Change rule	Minor: x.x; Major: x.0	Locked	Major change requires architecture review

2. Purpose & Positioning

TERAS Academy adalah sistem pembelajaran dan pengembangan kapabilitas yang dirancang untuk mengubah pengetahuan menjadi kemampuan yang dapat diterapkan, diamati, dibuktikan, dan ditingkatkan.

- Academy bukan sekadar penyedia materi pelatihan.
- Learning outcome harus diterjemahkan menjadi bukti kemampuan.
- Pembelajaran dapat berlangsung melalui self-learning/video, online, classroom, practical, atau blended.
- Untuk jalur pra-sertifikasi, Academy menyiapkan peserta; sertifikasi kompetensi tetap berada pada LSP/BNSP atau skema eksternal yang berlaku.

3. Master Architecture

Struktur TERAS Academy terdiri atas delapan lapisan yang saling terhubung:

- I. Strategic & Institutional Layer — mandat, positioning, target learner, capability framework, governance.
- II. Academy Learning Architecture — common foundation, learning architecture, curriculum architecture, modalities.
- III. Dual Pathway — Path A Capability Development dan Path B Pre-Certification.
- IV. Curriculum — program dan modul pembelajaran; baseline awal Solar PV M01–M10.
- V. Assessment & Evidence — knowledge, application, performance, evidence, gap, remediation, readiness.
- VI. Operational System — learner journey, SOP, forms, trainer/assessor, records.
- VII. Production System — module master, learner guide, theory, video, practical, assessment, evidence, remediation, trainer guide.
- VIII. Quality & Continuous Improvement — QA, KPI, learning loop, version control.

4. Common Foundation

Kedua pathway menggunakan fondasi Academy yang sama. Namun kesamaan fondasi tidak berarti kesamaan tujuan maupun assessment.

Prinsip kunci: Dua jalur memakai fondasi Academy yang sama, tetapi tujuan, kurikulum, assessment, output, dan KPI-nya berbeda.

- Common learning foundation dapat digunakan bersama.
- Evidence dan assessment dibedakan menurut tujuan pathway.
- Safety merupakan aturan lintas pathway.
- Tidak ada final achievement tanpa evidence yang memadai.

5. Dual Pathway

Aspek	Path A — Capability	Path B — Pre-Certification	Pembeda utama	Output
Tujuan	Membangun kapabilitas	Menyiapkan menuju assessment	Outcome akhir	Capability / readiness

		sertifikasi		
Kurikulum	Capability-oriented	Capability + competency mapping	Tambahan mapping	Target scheme
Assessment	Performance-oriented	Performance + scheme readiness	Basis penilaian	Internal vs external
Evidence	Capability evidence	Certification-oriented evidence	Kelengkapan bukti	Portfolio
KPI	Capability achievement	Gap closure & readiness	Ukuran keberhasilan	Outcome

6. Learning Architecture

- Self-learning / video tutorial: membangun pemahaman awal dan kesiapan teori.
- Online: interaksi, latihan, review, dan penguatan.
- Classroom: konseptualisasi, diskusi, demonstrasi, dan problem solving.
- Practical: demonstrasi dan pembuktian kemampuan.
- Blended: kombinasi sesuai learning outcome.

Learning principle: Theory modality may vary; learning outcome may not.

Practical learning pattern: Demonstrate → Guided Practice → Independent Practice → Feedback → Repeat/Re-practice.

7. Solar PV Curriculum Baseline

Kode	Modul	Fokus	Evidence
M01	PV Fundamentals	Sistem PV & aliran energi	E01 PV System Understanding
M02	Electrical Behavior, Measurement &	Pengukuran & interpretasi	E02 Measurement Record

	Interpretation		
M03	Safety & Work Practice	Keselamatan kerja	E03 Safety Observation Record
M04	PV System Architecture	Relasi sistem	E04 System Architecture Map
M05	Drawing & Work Preparation	Persiapan pekerjaan	E05 Work Preparation Record
M06	Mechanical Installation	Instalasi mekanikal	E06 Mechanical Installation Record
M07	Electrical Installation	Instalasi elektrikal	E07 Electrical Installation Record
M08	System Configuration & Integration	Integrasi sistem	E08 Integration Record
M09	Testing & Commissioning	Pengujian & commissioning	E09 Commissioning Record
M10	O&M & Basic Troubleshooting	Operasi, pemeliharaan & troubleshooting	E10 O&M/Troubleshooting Record

8. Assessment Architecture

Assessment menggunakan tiga lapisan:

- Knowledge — apakah peserta memahami konsep, prinsip, istilah, dan aturan.
- Application — apakah peserta dapat menerapkan pengetahuan pada situasi/tugas.
- Performance — apakah peserta dapat melakukan pekerjaan secara benar, aman, dan konsisten.

Rubric baseline: 0 Not demonstrated; 1 Requires substantial assistance; 2 Meets expected standard; 3 Strong/consistent performance.

Safety gate: kegagalan pada kriteria keselamatan kritis tidak dapat dikompensasikan oleh skor lain.

9. Evidence Architecture

Evidence harus memenuhi empat kualitas:

- Valid
- Relevant
- Sufficient
- Traceable

Evidence chain: Learning → Assessment → Evidence → Gap Analysis → Remediation → Re-assessment → Final Status.

Decision loop: Observe → Evidence → Diagnose → Decide → Act → Verify → Learn.

10. Gate Architecture

Gate	Nama	Makna
G1	Knowledge Ready	Pemahaman dasar memadai
G2	Safety Ready	Kesiapan kerja aman
G3	Practical Ready	Kesiapan melakukan tugas praktis
G4	System Ready	Kesiapan memahami/mengintegrasikan sistem
G5	Verification Ready	Kesiapan menguji dan menafsirkan hasil
G6	Operational / Pathway Ready	Kesiapan operasi atau masuk pathway berikutnya

11. Learner Journey

- Registration → Diagnostic Assessment → Pathway Assignment → Individual Learning Plan.
- Learning Delivery → Practical Training → Internal Assessment.
- Evidence Management → Status Decision.
- Jika belum tercapai: Gap Analysis → Remediation → Re-assessment.
- Path A: Capability Portfolio → Capability Achievement.
- Path B: Competency Mapping → Gap Closure → Mock Assessment → Readiness Assessment → Certification Readiness Portfolio → External Assessment.

12. Operational System & SOP

SOP-01 — Participant Registration

SOP-02 — Diagnostic Assessment

SOP-03 — Pathway Assignment

SOP-04 — Learning Delivery

SOP-05 — Practical Training

SOP-06 — Internal Assessment

SOP-07 — Evidence Management

SOP-08 — Remediation & Reassessment

SOP-09 — Certification Preparation

SOP-10 — External Certification Interface

SOP-11 — Trainer Management

SOP-12 — Assessor Management

SOP-13 — Quality Assurance

SOP-14 — Data & Record Management

SOP-15 — Version Control

Format SOP standar: Trigger → Input → Process → Decision → Output → Record → Owner.

13. Forms & Records

F-01 — Participant Profile

F-02 — Diagnostic Record

F-03 — Individual Learning Plan

F-04 — Practical Observation

F-05 — Assessment Record

F-06 — Evidence Record

F-07 — Remediation Plan

F-08 — Reassessment Record

F-09 — Capability Portfolio

F-10 — Certification Readiness Portfolio

F-11 — Trainer Report

F-12 — QA Report

Relationship: F01 → F02 → F03 → F04 → F05 → F06 → F09/F10; if not achieved → F07 → F08 → F05/F06.

14. Production System

- 01 Module Master
- 02 Learning Outcome
- 03 Learner Guide
- 04 Theory Content
- 05 Video/Tutorial

- 06 Classroom Guide
- 07 Practical Guide
- 08 Assessment Pack
- 09 Evidence Pack
- 10 Remediation Pack
- 11 Trainer Guide
- 12 QA Checklist
- 13 Version Record

Production principle: Build the learning once; differentiate the evidence and assessment according to the pathway.

15. Trainer & Assessor System

- Trainer bertanggung jawab atas delivery, demonstration, coaching, feedback, dan learning support.
- Assessor bertanggung jawab atas penilaian dan keputusan assessment.
- Trainer ≠ assessor secara fungsi. Jika satu orang menjalankan dua fungsi pada pilot, harus ada kontrol konflik kepentingan dan independent review.
- Assessment decision harus traceable terhadap criteria dan evidence.

16. Quality Assurance & Continuous Improvement

QA mencakup:

- Content QA
- Delivery QA
- Assessment QA
- Evidence QA
- Outcome/KPI QA
- Version & Change QA

Improvement loop: Assessment → Result → Evidence → Gap Analysis → Remediation → Re-assessment → Final Status → KPI → QA → Curriculum Improvement.

17. KPI Architecture

Pathway	KPI utama	Evidence
Path A	Capability achievement; practical performance; workplace application	Capability Portfolio / workplace evidence
Path B	Competency gap closure; mock assessment; readiness rate; external assessment participation; certification outcome	Readiness Portfolio / external result

18. Certification Boundary

TERAS Academy tidak menerbitkan sertifikasi kompetensi eksternal atas nama LSP/BNSP. Path B merupakan jalur persiapan: memetakan kompetensi terhadap skema target, mengidentifikasi gap, menutup gap, menyediakan evidence, dan menguji readiness. Keputusan sertifikasi berada pada lembaga sertifikasi dan skema yang berlaku.

19. Data, Records & Version Control

- One learner → one longitudinal record.
- Digital-first, printable when necessary.
- Minimum traceability fields: Learner ID, Module ID, Assessment ID, Evidence ID, Trainer/Assessor ID, Date, Result, Status.
- Minor change: v1.1; major architecture change: v2.0.
- Released documents must carry document ID, version, approval/status, and change record.

20. Production & Implementation Roadmap

- Checkpoint 1 — Master Architecture v1.0.
- Checkpoint 2 — Master Production Pack Template v1.0.
- Checkpoint 3 — M01 Golden Template.
- Checkpoint 4 — M02–M10 production in controlled waves.

- Wave 1: M01–M04 Foundation.
- Wave 2: M05–M07 Preparation & Installation.
- Wave 3: M08–M09 Integration & Verification.
- Wave 4: M10 Operation.
- Pilot → QA → Revision → Release.

21. Baseline / Locked Register

Baseline	Version	Status
TERAS Academy Master Architecture	v1.0	LOCKED
Dual Pathway Architecture	v1.0	LOCKED
Solar PV Curriculum M01–M10	v1.0	LOCKED
Assessment & Evidence Matrix	v1.0	LOCKED
SOP Architecture	v1.0	LOCKED
Master Forms & Records Pack	v1.0	LOCKED
Production Pack Architecture	v1.0	LOCKED

22. Master Design Rules

1. Common learning does not mean common assessment.

2. No final achievement without evidence.

3. Safety-critical failure is non-compensable.

4. Remediation is gap-specific, not generic module repetition.

5. Capability and certification are related but not interchangeable.

6. Evidence must be valid, relevant, sufficient, and traceable.

7. Every operational decision must leave a record.

8. Every released asset must have version control.

9. Build the learning once; differentiate evidence and assessment by pathway.

10. Architecture changes require controlled review.